

## 1. Executive Summary

|                         |        | Traffic Summary        |  | Count       |  |
|-------------------------|--------|------------------------|--|-------------|--|
| Metric                  | Value  | Total Flows Observed   |  | 99584       |  |
| Total Threats Mitigated | 19     | Total Packets Analyzed |  | 11513639817 |  |
| High Priority           | 1      | Malicious Packets      |  | 11488078697 |  |
| Low Priority            | 18     | True Negatives Passed  |  | 92846       |  |
| Manual Releases         | 0      |                        |  |             |  |
| Manual Blocks           | 0      | Attack Vector          |  | Count       |  |
|                         |        | ICMP Flood             |  | 14          |  |
| ML Metrics              | Value  | SYN Flood              |  | 0           |  |
| False Positives         | 13     | UDP Flood              |  | 0           |  |
| FP Rate                 | 0.01%  | Uncertain              |  | 5           |  |
| IF Precision            | 99.84% |                        |  |             |  |
| IF Recall               | 96.07% | Action                 |  | Count       |  |
| IF F1-Score             | 97.92% | Quarantined            |  | 3           |  |
| IF Accuracy             | 99.71% | Time Ban               |  | 3           |  |
|                         |        | Blackhole              |  | 3           |  |
|                         |        | Released               |  | 4           |  |

## 2. Performance Benchmark

### 2a. Isolation Forest: Anomaly Detection

| Metric                    | Value  | Description                                          |
|---------------------------|--------|------------------------------------------------------|
| Precision                 | 99.84% | Share of flagged anomalies that were genuine attacks |
| Recall (TPR)              | 96.07% | Share of actual attacks successfully flagged         |
| F1-Score                  | 97.92% | Balanced measure combining Precision and Recall      |
| Accuracy                  | 99.71% | Overall proportion of correct anomaly decisions      |
| False Positive Rate (FPR) | 0.01%  | Normal traffic incorrectly flagged as an attack      |
| False Negative Rate (FNR) | 3.93%  | Actual attacks that went undetected                  |
| True Positive Rate (TPR)  | 96.07% | Same measure as Recall, attacks correctly flagged    |
| True Negative Rate (TNR)  | 99.99% | Normal traffic correctly identified as safe          |

|                | Predicted: Attack | Predicted: Normal |
|----------------|-------------------|-------------------|
| Actual: Attack | TP 6693           | FN 274            |
| Actual: Normal | FP 11             | TN 92572          |

### 2b. Random Forest: Attack Classification

| Metric    | Value   | Description                                                  |
|-----------|---------|--------------------------------------------------------------|
| Precision | 100.00% | Share of classified flows assigned the correct attack type   |
| Recall    | 100.00% | Share of attacks of each type that were correctly identified |
| F1-Score  | 100.00% | Balanced measure combining Precision and Recall              |
| Accuracy  | 100.00% | Overall proportion of correct classifications                |

|           | Predicted: SYN | Predicted: ICMP | Predicted: UDP |
|-----------|----------------|-----------------|----------------|
| Act: SYN  | 0              | 0               | 0              |
| Act: ICMP | 0              | 6693            | 0              |
| Act: UDP  | 0              | 0               | 0              |

### 2c. Response Latency

| Metric                   | Value    | Description                                                             |
|--------------------------|----------|-------------------------------------------------------------------------|
| Detection Time           | 448.8 ms | Average time taken to detect an attack                                  |
| Mitigation Response Time | 6.0 ms   | Interval between the anomaly flag and the FlowMod blocking rule install |

2d. Controller Resource Overhead

| Metric                         | Value | Description                                                           |
|--------------------------------|-------|-----------------------------------------------------------------------|
| Controller CPU (Baseline)      | 1.16% | Ryu controller CPU usage under normal traffic                         |
| Controller CPU (Active Attack) | 2.74% | Ryu controller average CPU usage during a DDoS simulation             |
| Controller CPU (Mitigation)    | 2.83% | Ryu controller CPU usage during simultaneous detection and mitigation |

3. Offences Summary

| Source IP | Sessions | Max Offences | First Seen          | Last Seen           |
|-----------|----------|--------------|---------------------|---------------------|
| 10.0.0.23 | 9        | 4            | 2026-09-14 11:14:33 | 2026-09-14 11:36:17 |
| 10.0.0.24 | 9        | 4            | 2026-09-14 11:14:32 | 2026-09-14 11:36:15 |
| 10.0.0.25 | 8        | 2            | 2026-09-14 11:14:33 | 2026-09-14 11:36:17 |
| 10.0.0.1  | 1        | 1            | 2026-09-14 11:31:20 | 2026-09-14 11:31:30 |
| 10.0.0.5  | 1        | 1            | 2026-09-14 11:21:18 | 2026-09-14 11:21:28 |

#### 4. Chronological Mitigation Log

| Timestamp           | Source IP | Class | Vector     | Confidence | Priority | Action        |
|---------------------|-----------|-------|------------|------------|----------|---------------|
| 2026-09-14 11:14:32 | 10.0.0.24 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-14 11:14:33 | 10.0.0.25 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-14 11:14:33 | 10.0.0.23 | DDoS  | Uncertain  | -          | Low      | Quarantined   |
| 2026-09-14 11:14:43 | 10.0.0.24 | DDoS  | ICMP Flood | 91.0%      | Critical | Time Ban (1m) |
| 2026-09-14 11:14:43 | 10.0.0.24 | DDoS  | ICMP Flood | 91.0%      | Critical | Time Ban      |
| 2026-09-14 11:14:44 | 10.0.0.25 | DDoS  | ICMP Flood | 91.2%      | Critical | Time Ban (1m) |
| 2026-09-14 11:14:44 | 10.0.0.23 | DDoS  | ICMP Flood | 91.0%      | High     | Time Ban (1m) |
| 2026-09-14 11:14:44 | 10.0.0.25 | DDoS  | ICMP Flood | 91.2%      | Critical | Time Ban      |
| 2026-09-14 11:14:44 | 10.0.0.23 | DDoS  | ICMP Flood | 91.0%      | Critical | Time Ban      |
| 2026-09-14 11:15:43 | 10.0.0.24 | DDoS  | ICMP Flood | 92.1%      | Critical | Time Ban (2m) |
| 2026-09-14 11:15:44 | 10.0.0.25 | DDoS  | ICMP Flood | 94.7%      | Critical | Time Ban (2m) |
| 2026-09-14 11:15:44 | 10.0.0.23 | DDoS  | ICMP Flood | 93.2%      | Critical | Time Ban (2m) |
| 2026-09-14 11:21:28 | 10.0.0.5  | DDoS  | Uncertain  | 59.5%      | Low      | Released      |
| 2026-09-14 11:21:43 | 10.0.0.24 | DDoS  | ICMP Flood | 92.1%      | Critical | Released      |
| 2026-09-14 11:21:43 | 10.0.0.23 | DDoS  | ICMP Flood | 94.2%      | Critical | Released      |
| 2026-09-14 11:21:55 | 10.0.0.24 | DDoS  | ICMP Flood | 92.1%      | Critical | Blackhole     |
| 2026-09-14 11:21:55 | 10.0.0.23 | DDoS  | ICMP Flood | 94.2%      | Critical | Blackhole     |
| 2026-09-14 11:25:58 | 10.0.0.25 | DDoS  | ICMP Flood | 91.2%      | Critical | Blackhole     |
| 2026-09-14 11:31:30 | 10.0.0.1  | DDoS  | Uncertain  | 58.9%      | Low      | Released      |

#### 5. IP Attack History (Completed Sessions)

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.25 | ICMP Flood | 0.6462   | 83.3% | Critical | Phase 3 | 1        | 2s       | 2026-09-14 11:36:17 | Escalated to Blackhole                    |
| 10.0.0.23 | ICMP Flood | 0.6460   | 83.9% | Critical | Phase 3 | 1        | 2s       | 2026-09-14 11:36:17 | Escalated to Blackhole                    |
| 10.0.0.24 | ICMP Flood | 0.6462   | 82.9% | Critical | Phase 3 | 1        | 1s       | 2026-09-14 11:36:15 | Escalated to Blackhole                    |
| 10.0.0.1  | Uncertain  | 0.6080   | 58.9% | Low      | Phase 1 | 1        | 10s      | 2026-09-14 11:31:30 | Attack Stopped                            |
| 10.0.0.24 | ICMP Flood | 0.6447   | 90.0% | Critical | Phase 3 | 1        | 2s       | 2026-09-14 11:31:02 | Escalated to Blackhole                    |
| 10.0.0.23 | ICMP Flood | 0.6458   | 83.3% | Critical | Phase 3 | 1        | 1s       | 2026-09-14 11:31:02 | Escalated to Blackhole                    |
| 10.0.0.25 | ICMP Flood | 0.6456   | 83.5% | Critical | Phase 3 | 1        | 1s       | 2026-09-14 11:31:02 | Escalated to Blackhole                    |
| 10.0.0.25 | ICMP Flood | 0.6461   | 91.2% | Critical | Phase 2 | 1        | 11s      | 2026-09-14 11:25:58 | Attack persisted - escalating to time ban |

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.25 | ICMP Flood | 0.6317   | 91.2% | Critical | Phase 3 | 2        | 11s      | 2026-09-14 11:25:58 | Escalated to Blackhole                    |
| 10.0.0.24 | ICMP Flood | 0.6459   | 83.9% | Critical | Phase 3 | 1        | 2s       | 2026-09-14 11:25:48 | Escalated to Blackhole                    |
| 10.0.0.23 | ICMP Flood | 0.6496   | 79.8% | Critical | Phase 3 | 1        | 1s       | 2026-09-14 11:25:48 | Escalated to Blackhole                    |
| 10.0.0.24 | ICMP Flood | 0.6305   | 92.1% | Critical | Phase 3 | 4        | 11s      | 2026-09-14 11:21:55 | Escalated to Blackhole                    |
| 10.0.0.23 | ICMP Flood | 0.6458   | 94.2% | Critical | Phase 3 | 4        | 11s      | 2026-09-14 11:21:55 | Escalated to Blackhole                    |
| 10.0.0.24 | ICMP Flood | 0.6335   | 92.1% | Critical | Phase 2 | 3        | 10s      | 2026-09-14 11:21:54 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6352   | 94.2% | Critical | Phase 2 | 3        | 10s      | 2026-09-14 11:21:54 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6410   | 92.1% | Critical | Phase 2 | 1        | 1m 11s   | 2026-09-14 11:21:43 | Ban Expired                               |
| 10.0.0.25 | ICMP Flood | 0.6416   | 92.6% | Critical | Phase 2 | 2        | 1m 10s   | 2026-09-14 11:21:43 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6313   | 94.2% | Critical | Phase 2 | 1        | 1m 10s   | 2026-09-14 11:21:43 | Ban Expired                               |
| 10.0.0.5  | Uncertain  | 0.5869   | 59.5% | Low      | Phase 1 | 1        | 10s      | 2026-09-14 11:21:28 | Attack Stopped                            |
| 10.0.0.24 | ICMP Flood | 0.6464   | 91.2% | Critical | Phase 2 | 1        | 11s      | 2026-09-14 11:20:43 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6454   | 91.2% | Critical | Phase 2 | 1        | 10s      | 2026-09-14 11:20:43 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6445   | 90.9% | Critical | Phase 2 | 1        | 10s      | 2026-09-14 11:20:43 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6223   | 94.7% | Critical | Phase 2 | 2        | 1m 11s   | 2026-09-14 11:15:44 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6413   | 93.2% | Critical | Phase 2 | 2        | 1m 11s   | 2026-09-14 11:15:44 | Attack persisted - escalating to time ban |
| 10.0.0.24 | ICMP Flood | 0.6417   | 92.1% | Critical | Phase 2 | 2        | 1m 11s   | 2026-09-14 11:15:43 | Attack persisted - escalating to time ban |
| 10.0.0.25 | ICMP Flood | 0.6454   | 91.2% | Critical | Phase 2 | 1        | 11s      | 2026-09-14 11:14:44 | Attack persisted - escalating to time ban |
| 10.0.0.23 | ICMP Flood | 0.6445   | 91.0% | High     | Phase 2 | 1        | 11s      | 2026-09-14 11:14:44 | Attack persisted - escalating to time ban |

| Source IP | Vector     | IF Score | Conf  | Priority | Phase   | Offences | Duration | Unblocked At        | Reason                                    |
|-----------|------------|----------|-------|----------|---------|----------|----------|---------------------|-------------------------------------------|
| 10.0.0.24 | ICMP Flood | 0.6455   | 91.0% | Critical | Phase 2 | 1        | 11s      | 2026-09-14 11:14:43 | Attack persisted - escalating to time ban |

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE